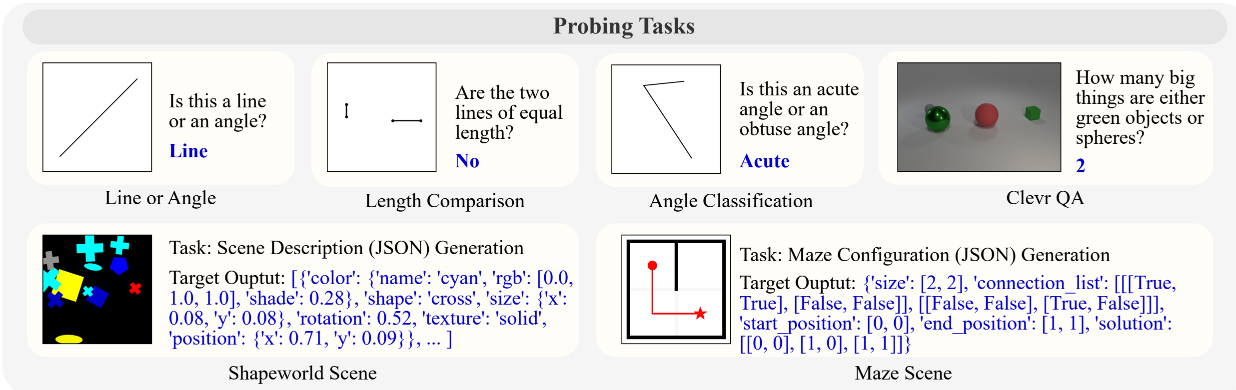

Figure 8: Illustration of the probing tasks. The four tasks at the top are question-answering tasks, while the two tasks at the bottom are scene-generation tasks. The goal of the scene-generation tasks is to generate the entire structured scene description following a predefined schema.

		Input Type	Line or Angle	Angle Classification	Length Comparison	Clevr QA
Zero-Shot	GPT-4V	Image	1.00	0.58	0.64	0.57
	GPT-4	SVG	0.45	0.47	0.60	0.36
Finetuned	Llava-v1.5-7b	Image	0.50	0.50	0.50	0.45
	Vicuna	SVG	0.93	0.70	0.99	0.54

		Input Type	Shapeworld Scene		Maze Scene		
			shape (acc↑)	position (l2↓)	connectivity (acc↑)	start-pos (acc↑)	end-pos (acc↑)
Zero-Shot	GPT-4V	Image	0.33	0.27	0.27	0.21	0.22
Finetuned	Llava-v1.5-7b	Image	0.04	0.67	0.26	0.03	0.03
	Vicuna	SVG	0.15	0.07	0.54	0.08	0.09

Table 3: Probing task results. We report the accuracy for the four question-answering tasks at the top. At the bottom, we use different metrics for different fields in the predicted scene description JSON. "acc" refers to accuracy (larger is better) while "l2" refers to the Euclidean distance between the predicted and ground truth [x, y] coordinates (lower is better). Scores with a blue background denote the better fine-tuned method compared to the SVG and Image representation. Scores with a red background denote tasks where fine-tuned methods cannot outperform zero-shot GPT-4V. Detailed analysis can be found in § A.1.

A.2 Llava's Failure Mode in Visual Reasoning with Vector Graphics

We further investigate whether the difficulty in understanding low-level visual features of Llava models stems from (1) the visual backbone itself, i.e., CLIP, or (2) the bridge between the visual backbone and the LLM backbone. We include a set of **Linear Probing** experiments on three binary classification probing tasks, where we train a simple linear classifier based on the visual backbone features (before and after projection) of the Llava model. As shown in Figure 9:

(1) In tasks requiring more precise low-level perception, such as Angle Classification and Length Comparison, CLIP embeddings are inherently less effective at capturing relevant features. Furthermore, as shown in Figure 10, in some tasks, e.g., Length Comparison, linear regression even fails to achieve 90%+ training accuracy after 10 epochs of training, struggling to converge.

(2) When connected to an LLM using the projection layer, the visual features in Llava become less effective for low-level visual reasoning. Additionally, there is a significant gap between linear probing and instruction fine-tuning performance. These results suggest that even if the backbone does preserve useful features, the LLM cannot effectively leverage those visual tokens after projection.

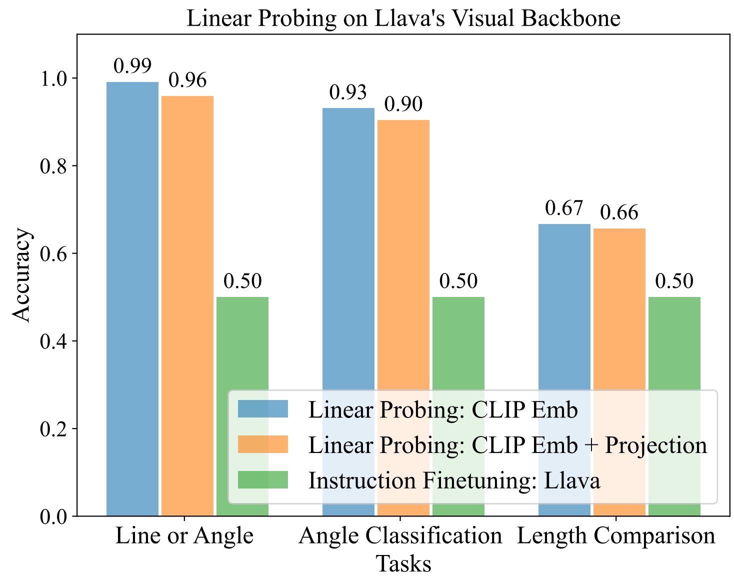

Figure 9: The average accuracy of linear probing, computed across ten epochs. Detailed training and testing scores for each epoch can be found in Figure 10. The results demonstrate that (1) CLIP embeddings are less effective for tasks requiring precise perception, such as Length Comparison, in comparison to tasks that emphasize on higher-level semantics, such as Line or Angle; (2) connecting to an LLM through the widely-used Llava-style architecture results in further diminished performance on tasks involving low-level visual details.

We hypothesize that the failure mode likely stems from the multimodal pretraining and instruction-tuning paradigm, where the tasks are biased towards high-level semantics, such as image captioning (Lin et al., 2014; Sidorov et al., 2020) and natural-image-based VQA (Goyal et al., 2017; Krishna et al., 2017; Marino et al., 2019; Schwenk et al., 2022). The training mixtures (Liu et al., 2023b;a; Dai et al., 2023; Chen et al., 2023a) for current LMMs predominantly focus on high-level features of images, providing little incentive for models to retain low-level visual details. For example, the caption of an image containing a 2D maze, such as the one shown in Figure 2, is likely to be "A 2×2 maze with black lines, a red circle and a star." and may not include detailed configurations of the mazes, such as the precise locations of the walls, the red circle, and the red star.

A.3 Remaining Challenges of Using SVG Representations

Although we have demonstrated that SVG can serve as a promising alternative representation for reasoning about vector graphics, we identify several remaining challenges:

(1) Pretrained LLMs, including the most capable ones such as GPT-4 (OpenAI, 2023a), possess limited out-of-the-box understanding of SVG code. This limitation is evidenced by the low zero-shot performance of GPT-4 with SVG input (see row 2 in Table 3).

(2) Even after finetuning, the SVG-based LLM may still underperform zero-shot GPT-4V on certain tasks, particularly those involving complex scenes, such as Shapeworld Scene and Maze Scene. In these instances, the SVG code becomes excessively verbose. These findings suggest that learning a model to directly comprehend the raw SVG code of an entire image poses significant challenges.

(3) A fundamental challenge, irrespective of the chosen representation for visual input, is the lack of generalization capability to unseen tasks and various vector graphics image domains. If we rely on existing LMM training mixtures, even any image can be converted into SVG code, the tasks remain biased towards high-level semantics. In addition, it is infeasible to directly manually construct and annotate ⟨SVG, question, answer⟩ pairs covering diverse tasks with vector graphics.

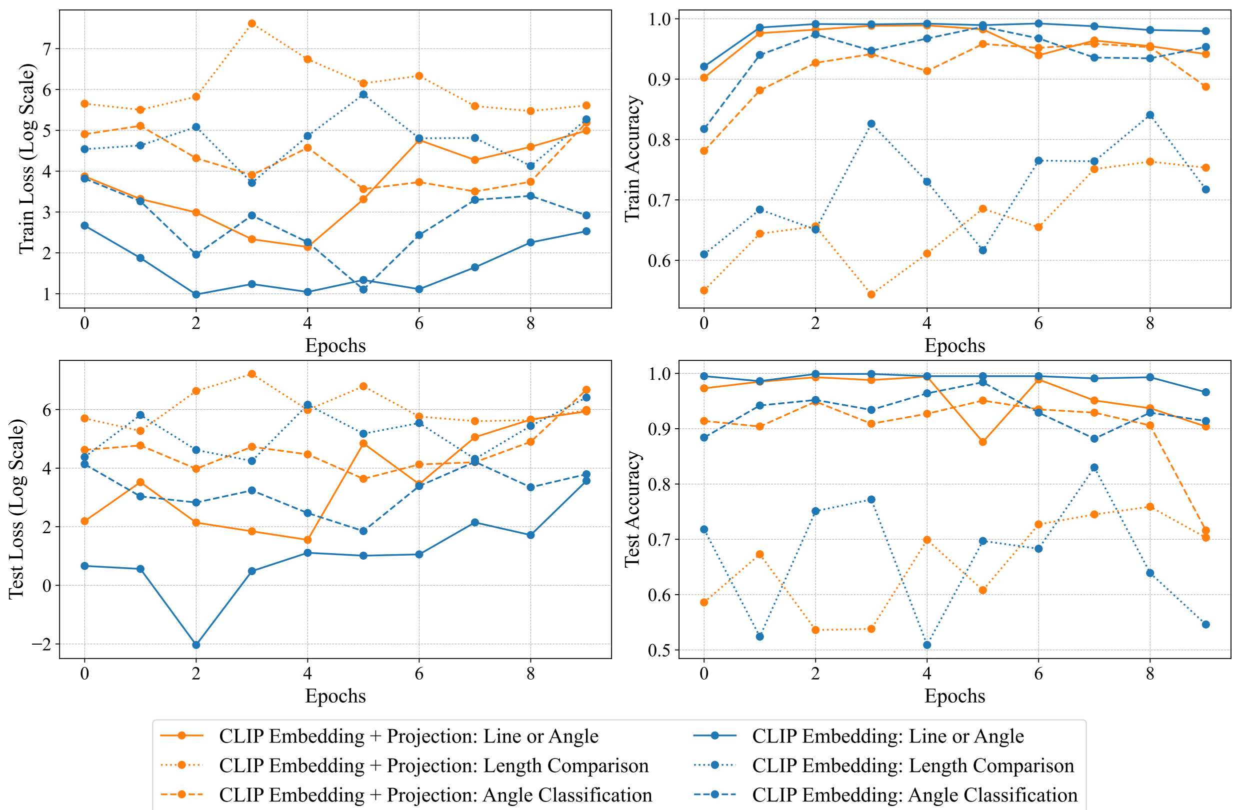

Figure 10: Linear probing training details: Different line styles represent different tasks, while different colors refer to different visual embeddings used for training the linear classifier. The training loss (top-left) shows that the projected embedding (orange lines) learns at a slower pace compared to the original CLIP embedding (blue lines). The training accuracy (top-right) reveals that for certain tasks, such as Length Comparison, the model continues to struggle with overfitting the training set even after 10 epochs.

These challenges motivated us to propose another layer of abstraction, the Primal Visual Description, aimed at bridging the gap between low-level perception and high-level language reasoning on downstream tasks.

B Error Analysis Details

As introduced in § 2, the proposed VDLM consists of two stages focused on perception—namely, Image-to-SVG and SVG-to-PVD, and one stage focused on reasoning, i.e., PVD-to-final answer. We aim to investigate the errors in both the perception and reasoning modules.

For each task, we manually examine 10 error cases and determine whether the error primarily stems from the perception stage or the reasoning stage. We task a human with reviewing the reconstructed image from the PVD representation and assessing the question of the task instance. If, for a human, the reconstructed image is still insufficient for solving the task, we classify this error as a perception error. Otherwise, it is categorized as a reasoning error. Figure 11 illustrates the distribution of errors between perception and reasoning stages. We further identify some typical categories of perception and reasoning errors as follows:

Common perception errors. (1) **Novel shapes not covered by the Primal Visual Description (PVD)**: For example, as visualized in Figure 12, the Shapeworld dataset includes a "semicircle" shape type which is not in the PVD ontology; we see that the learned SVG-to-PVD model tends to predict it as an ellipse. This perception error directly contributes to the inferior performance of VDLM-mm compared to GPT-4o on the Shapeworld tasks, as shown in Table 1.